

Development:Phase-2/Educational Intelligence

Thursday, 28 May 2026 2:09 PM

(Science Project + Teacher Assistant)

This roadmap starts AFTER:

Phase-1.1 completed

Meaning:

- LLM integration already done
- story/activity generation working
- prompt flow basics available

Now we evolve:

from generic AI generation

to

educational intelligence platform

STEP-1 — Create New Domain Modules

Goal

Separate new educational domains cleanly.

Create

```
services/  
├── science_projects/  
├── teacher_assistant/  
├── orchestration/  
└── validations/
```

Also:

```
prompts/  
├── science_projects/  
└── teacher_assistant/
```

STEP-2 — Create Science Project Prompt Templates

Goal

Generate structured science projects.

Add Prompts For

- project generation
- material list generation
- build steps
- explanation generation
- difficulty estimation

Output

Consistent project JSON response structure.

STEP-3 — Create Teacher Assistant Prompt Templates

Goal

Generate educational assessments.

Add Prompts For

- MCQ generation
- short answer generation
- long answer generation
- worksheet generation
- exam-paper generation

STEP-4 — Define Educational Schemas

Goal

Move from free text → structured educational outputs.

Create

.

```
schemas/
├── science_project.py
├── material_list.py
├── assessment.py
├── worksheet.py
```

● STEP-5 — Create Educational Metadata Models

Goal

Introduce curriculum awareness.

Create

```
models/
├── curriculum.py
├── project_metadata.py
├── assessment_pattern.py
```

Store:

- class
- subject
- chapter
- topic
- difficulty
- estimated cost
- estimated build time

● STEP-6 — Build Science Project Service

Goal

Core project generation engine.

Responsibilities

- topic validation
- prompt orchestration
- project generation
- output formatting
- metadata injection

Example

Class 6

Topic: Water Conservation

→ returns:

- project idea
- materials
- workflow
- explanation

● STEP-7 — Build Teacher Assistant Service

Goal

Core assessment generation engine.

Responsibilities

- chapter-based question generation
- exam-pattern generation
- worksheet generation
- marks distribution handling

● STEP-8 — Create Workflow Pipelines

Goal

Standardize AI workflows.

Create

```
science_project_pipeline.py
teacher_assessment_pipeline.py
```

Each pipeline handles:

- validation
- metadata enrichment
- prompt execution
- formatting

● STEP-9 — Create Prompt Orchestration Layer

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Goal

Centralize LLM workflow handling.

Create

services/orchestration/

Responsibilities:

- model routing
- context injection
- retry handling
- response cleanup

● STEP-10 — Create Provider Abstraction

Goal

Support multiple models cleanly.

Create

core/providers/

```
|— base_provider.py
|— openai_provider.py
|— gemini_provider.py
|— ollama_provider.py
```

● STEP-11 — Add Structured API Endpoints

Create APIs

Science Projects

/api/v1/science-projects/generate

Teacher Assistant

/api/v1/teacher-assistant/generate

● STEP-12 — Introduce Validation Layer

Goal

Prevent bad educational outputs.

Add Checks For

- missing materials
- invalid grade difficulty
- malformed question papers
- unrealistic project complexity

● STEP-13 — Add Educational Configurations

Goal

Avoid hardcoding.

Add Configs

config/

```
|— grade_rules.py
|— difficulty_rules.py
|— assessment_patterns.py
```

● STEP-14 — Add Initial Curriculum Metadata

Goal

Prepare for future intelligence layer.

Start Small

Add:

- few classes
- few science chapters
- few assessment patterns

DO NOT ingest full NCERT yet.

● STEP-15 — Add Project Recommendation Logic

Goal

Generate better project suggestions.

Logic Inputs

- grade
- topic
- difficulty
- material complexity
- estimated budget

● STEP-16 — Add Teacher Exam Pattern Logic

Goal

Support realistic exam workflows.

Support

- MCQ counts
- short/long ratio
- marks distribution
- difficulty balancing

● STEP-17 — Create Prototype Instruction Output

Goal

Support physical model builder.

Output Must Include

- materials
- quantities
- build steps
- estimated cost
- precautions

● STEP-18 — Add Documentation Support APIs

Goal

Store prototype metadata later.

Prepare APIs For

- photos
- notes
- build records
- cost sheets

Simple local storage initially is enough.

● STEP-19 — Create Initial Demo Dataset

Goal

Create reusable educational examples.

Prepare

- 10 science topics
- 10 project examples
- 5 assessment examples

This helps future testing.

● STEP-20 — Manual Educational Validation Phase

Goal

You manually review outputs first.

Check:

- educational quality
- practicality
- feasibility
- project clarity

Do NOT automate too early.

● STEP-21 — Build Basic Regression Foundation (LAST)

Now only start:

toilage-regression-lab

Add:

- API regression

- prompt regression
- golden output testing

Keep minimal initially.



MOST IMPORTANT ADVICE

During this entire phase:

prioritize output quality over architecture perfection

Your biggest risk now is:

building infrastructure instead of educational value

Focus on:

- useful projects
- practical workflows
- strong teacher outputs
- prototype feasibility

That creates real value fastest.